

Movie Booking Conversational AI

Group 151 - Executed Notebook Results Summary

This PDF summarizes the executed notebook, including encoder training, comparative LLM prompting results, tokenizer analysis, OOD evaluation, and key conclusions required by the rubric.

Key Findings

- ? Encoder intent accuracy: 0.987
- ? Encoder entity strict F1: 0.643
- ? Structured JSON LLM intent accuracy: 0.546
- ? Tokenizer OOV (basic vs BPE test): 0.0196 vs 0.0102
- ? Adversarial encoder intent accuracy: 0.200
- ? Adversarial LLM intent accuracy: 0.050

Tokenizer OOV Comparison

Tokenizer	Validation OOV	Test OOV
Basic word tokenizer	0.0193	0.0196
Custom BPE tokenizer	0.0066	0.0102
BERT WordPiece	0.0000	0.0000

LLM Strategy Comparison

Strategy	Intent Acc.	Macro F1	Strict F1	Partial F1	Mean Latency (ms)	Cost / Query	Cost / 1000
zero_shot	0.408	0.401	0.801	0.843	0.124	0.001044	1.0440
few_shot	0.526	0.503	0.863	0.907	0.106	0.002651	2.6509
structured_json	0.546	0.522	0.867	0.911	0.098	0.001290	1.2902

Encoder vs LLM Trade-off Summary

- ? Encoder-only models remain relevant because they are smaller, faster, deterministic, and easier to deploy on-premise.
- ? LLM prompting is useful when zero-shot generalization or flexible reasoning is needed, but it is costlier and less controllable.
- ? For movie-booking NLU, the notebook shows the encoder is the stronger production choice when latency, privacy, and reproducibility matter.

Dataset and Model Notes

- ? Dataset size: 1000 examples split into train=704, val=144, test=152.
- ? Tokenizer vocab size: 336.
- ? Encoder model size: 0.345 MB.
- ? The notebook also includes adversarial/OOD evaluation, error analysis, visualizations, and a technical report in the repository.

Submission Status

? Executed notebook: included as Group_151.ipynb and exported HTML results.

? Annotation schema documentation: included.

? Technical report: included.

? ZIP package: regenerated after the notebook updates.

? PDF artifact: generated as a summary report because nbconvert PDF export is blocked by missing pandoc in this environment.

End of document.